



Randi Sims, Ph.D.

Postdoctoral Fellow
Clemson University

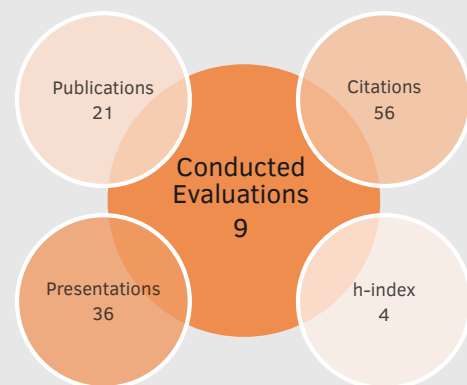
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Quality Evaluation Designs

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Profiles



Metrics



References

Dr. Karen High
Advisor – Clemson University
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Dr. Gary Lichtenstein
Founder – Quality Evaluation Designs
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Dr. Kristin Frady
Faculty – Clemson University
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Objective

Dr. Randi Sims is a mixed-methods evaluation and research scientist with a Ph.D. in Engineering and Science Education. She has six years of experience leading and supporting 9 NSF-funded evaluations, 3 research projects, and 2 outreach programs in K-12 and higher STEM education. Her expertise spans evaluation design, instrument development and validation, grant development, IRB-compliant data collection, quantitative and qualitative analysis, epistemic network analysis, and translating findings into actionable recommendations for stakeholders. She is the current Owner and Principal Evaluator at Quality Evaluation Designs, LLC and an externally-funded postdoctoral fellow at Clemson University.

Education

Postgraduate Studies

2022 – 2025 **Ph.D. in Engineering and Science Education** Clemson, SC
Title: Legitimacy, Learning, and LAUNCH: Exploring Relationships, Community, and Connection in an Undergraduate Research Experience

Ethnographic study of undergraduate lab experiences as communities of practice, using machine learning to validate coded discourse data.

2020 – 2022 **M.S. in Biological Sciences** Clemson, SC
Title: Evaluating the Impact of Research and Outreach Marine Science Programs on Elementary and Undergraduate Students
Quantitative study of knowledge, skill, and identity development in undergraduate marine science researchers, including a K-5 outreach initiative using citizen science and virtual reality.

Other Training

Fall 2023 **Data Science in Education with R Workgroup** Clemson, SC
Training in advanced descriptive, inferential, and predictive statistics using R, contextualized to education research and evaluation.

Skills

Evaluation

Program Evaluation

- 9 evaluations led or co-led
- 3 evaluations as principal evaluator
- Multi-million-dollar, multi-stakeholder projects

Instrument Development

- 4 validated surveys
- 10+ interview and focus group protocols
- 3 observation protocols
- EFA/CFA

Logic Models

- 4 logic models for federal submissions (2 funded)
- Co-facilitated logic model workshop

IRB Compliance

- Protocols developed across 5 institutions
- 10 total IRB protocols developed

Research & Analysis

Mixed Methods

- Thematic, discourse, and ML-assisted qualitative coding
- Predictive and inferential quantitative modeling

Quantitative Analysis

- R, Python, JMP, SPSS
- Excel, Google Sheets
- Network Analyses
- Linear regression

Qualitative Analysis

- 100+ interviews
- 4 focus groups
- 1 ethnographic study
- MAXQDA, NVivo

Literature Reviews

- 3 systematic reviews
- 1 publication in review
- 1 publication in prep

Impact & Leadership

Reporting & Dissemination

- Funder, practitioner, and scholarly reports
- Findings translated into actionable recommendations

Grant Development

- 5 national submissions
- 10+ state submissions
- 12 federal grant reviews
- Multi-million-dollar budgets

Project Management

- Program manager (2)
- Student manager (90+)
- Faculty mentor (6)

Team Management

- 5 employees managed (National Brick Research Center)
- 4 consultants managed (QED)

Selected Evaluation and Research Projects

Quality Evaluation Designs, LLC

These NSF-funded evaluations include supporting research and program funding submissions and leading evaluation design and implementation. Links are provided.

Ongoing	<u>Evaluating Mentoring Practices in STEM Graduate Student Education and Research</u> Role: Principal Evaluator <i>Designed, currently implementing and reporting internal evaluation on program broader impacts and intellectual merit.</i>
Ongoing	<u>Engineering Education and Research Centers 2024 & 2026 Conferences</u> Role: Principal Evaluator <i>Serving as lead evaluator to determine conference impacts on federally-supported engineering education efforts.</i>
Ongoing	<u>Rural Engineering Student Sense of Belonging at UW-Madison</u> Role: Research Consultant <i>Currently training new engineering education research faculty on qualitative research methods and survey development.</i>
Ongoing	<u>Concept Mapping for Improved Technical Communication in Undergraduates</u> Role: Research Consultant <i>Currently training new engineering education research faculty on thematically analyzing focus group data.</i>
Ongoing	<u>Student Research on Sweden's Energy Transition</u> Role: Principal Evaluator <i>Currently leading evaluation efforts to determine impact and program improvements for an undergraduate Swedish exchange program.</i>
2024 – 2026	<u>RIEF Virtual Communities of Practice</u> Role: Senior Consultant <i>Supported evaluation as lead data collector and analyst for a community of practice for new engineering education researchers.</i>
2025	<u>Penn State University's Acoustics Graduate Student Program</u> Role: Senior Consultant <i>Supported evaluation as lead data collector and analyst to support departmental transitions in Penn State's Acoustics program.</i>

Clemson University

Postdoctoral Fellow

Ongoing	<u>Rural Engineering Transfer Student Assets</u> <i>Postdoctoral research examining rural engineering transfer student assets using qualitative methods analyzed via supervised machine learning and Quantitative Ethnography. Supervised by Dr. Kris Frady.</i>
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Clemson University

Graduate Research Assistant

Oct 2024 – Dec 2025	<u>Elevating Mentoring Practices in Graduate Studies</u> <i>NSF-funded initiative to enhance holistic mentoring in STEM graduate education by promoting systemic change in mentorship practices, mental health support, and professional development. Contributed to grant development, budgeting, and submission, programming, and curriculum development. Led IRB development and submission, qualitative research methods and scholarly dissemination.</i>
2022 – 2025	<u>Valuing Education and Career Transition Opportunities</u> <i>NSF CAREER-funded research project on strengthening transfer pathways into engineering. Led qualitative interviews, survey development and validation, quantitative analyses. Supported development of digital transfer dashboard and scholarly dissemination. Now serving as lead evaluator and supporting program implementation.</i>
2022 – 2023	<u>Peer Review Training</u> <i>NSF EAGER-funded project aimed at improving equity and quality of publication review process through peer review mentor training. Supported evaluation activities, qualitative data collection and analysis, as well as scholarly dissemination.</i>

Selected Peer-Reviewed Publications

Journal Articles and Book Chapters

- **Sims, Randi.** (2025a). *Legitimacy, learning, and launch: Exploring relationships, community, and connection in an undergraduate research experience* [Manuscript in preparation]
- **Sims, Randi.** (2025b). *Science undergraduate research communities: A review of the literature* [Manuscript in preparation]
- Frady, K. K., & **Sims, Randi.** (2026). *Toward an engineering transfer student capital theory* [Manuscript under review at *Journal of Engineering Education*]
- Watts, K., **Sims, Randi**, Ko, E., Jensen, K., Bates, R., Lichtenstein, G., & Benson, L. (2023). Peer reviewer training to build capacity in engineering education research. *Australasian Journal of Engineering Education*, 1–22. <https://doi.org/10.1080/22054952.2023.2214459>
- **Sims, Randi**, Tallapragada, M., Payton, T. G., Noonan, K., Prosser, K. L., & Childress, M. J. (2021). University Experiences of Marine Science Research and Outreach Beyond the Classroom. *Integrative and Comparative Biology*, 61(3), 1078–1088. <https://doi.org/10.1093/icb/icab104>
- Tallapragada, M., **Sims, Randi J.**, Payton, T. G., Noonan, K. R., Bridgeford, K. E., Smith, K. M., Fuentes, M., Prosser, K. L., & Childress, M. J. (2023). Something Very Fishy (SVF): Avenues of Communication to Inspire Engagement with Climate and Marine Science. In J. G. Burchfield & A. A. Kedrowicz (Eds.), *Teaching communication across disciplines for professional development, civic engagement, and beyond*. Lexington Books

Conference Publications

- **Sims, Randi**, Watts, K., Stephan, A. T., Ta, T. N. Y., Sharpe, A., & Lichtenstein, G. (2025). Msi faculty on the rise: Strengthening federal grant proposals through cross-institution collaborations and networking [<https://peer.asee.org/56980>]. *Proceedings of the 2025 ASEE Annual Conference & Exposition*. <https://doi.org/10.18260/1-2--56980>
- Yeaman, A., Yuan, X., Lamas-Samanamud, G., Beem, H., Moore, J. M., & **Sims, Randi.** (2024). WIP: Survey validation to enable investigating community cultural wealth in engineering students' first year experiences (FYE). <https://doi.org/10.18260/1-2--48642>
- Frady, K. K., & **Sims, Randi.** (2023). A Qualitative Exploration of an Assets-Based Approach to Building Engineering Transfer Student Capital. *2023 IEEE Frontiers in Education Conference (FIE)*, 1–5. <https://doi.org/10.1109/FIE58773.2023.10343352>

Selected Research Grants and Scholarships

As an evaluator, I have supported 12 grant submissions currently under review. Details are omitted to protect project intellectual property.

Supported

\$1,000,000	NSF Innovations in Graduate Education: EMPOWERS	2024-Ongoing
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\$156,000	American Society for Engineering Education eFellowship	2026-Ongoing
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\$6,250	Graduate Student Travel Grants (7)	2020-2025
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Selected Presentations

Workshops

- **Sims, Randi** & Lichtenstein, G. (2026, June). *Unleash the power of the logic model* [QED LLC 30th Celebration] [Workshop]. Charlotte, NC
- High, K. A., Brown, J., & **Sims, Randi.** (2023, June). *Faculty Development for Culturally Responsive Mentoring of Graduate and Undergraduate Students in Research: Challenges and Strategies* [2023 ASEE Annual Conference & Exposition] [Workshop]. Baltimore, Maryland

Oral Presentations and Posters

- Sims, A., Magnin, N., Montalvo, M., & **Sims, Randi.** (2023, April). *What's In Our Waters? We Offer Answers!* [Focus on Creative Inquiry] [Poster Presentation]. Clemson, SC
- Ransom, T., Manning, J., & **Sims, Randi.** (2023, January). *A community of practice for STEM education graduate students: Lunch & Learn* [Southeastern STEM Education Research Conference] [Oral Presentation]. Cookeville, TN
- **Sims, Randi** & Childress, M. (2022, January). *Projects are not just for kids: Integrating science outreach projects into undergraduate learning during a pandemic* [Society for Integrative and Comparative Biology] [Oral Presentation]. Virtual

Review Duties

2026	Journal of Women & Minorities in Sci. & Eng.	Reviewer
2023 – 2025	ASEE Annual Conference	Reviewer
2022 – 2023	Journal of Biology Education	Reviewer
2023 – 2024	Frontiers in Education Conference	Reviewer